Attention: what changes when the input changes?

Two input tokens · one selected query · scalar projection weights W_Q = W_K = W_V = 1.

1 · Compute attention

Input: x = [1, 3] Reference: [0, 0]

Query q = 1 Keys K = [1, 3]

Values V = [1, 3] Head dimension = 1

$$s = qK = [1, 3]$$

$$A = \mathrm{softmax}(s)$$

0.1192	0.8808

Weights for token 1 and token 2

$$o = 0.1192 \times 1 + 0.8808 \times 3$$

Output o = 2.7616

2 · Separate two routes

$$do = A\,dV + (dA)V$$

At the clean input:

Value route gradient [0.1192, 0.8808]

Weight route gradient [0.2100, 0.2100]

Input path: x(α) = [α, 3α]

Both routes change along this path.

Integrate the two gradients, then

multiply by each input difference.

3 · Obtain an input map

Value route

0.3524	1.9427

+

Weight route

0.1166	0.3499

=

Full attention learning image

0.4691	2.2925

Same total, different input allocation: frozen clean A gives [0.1192, 2.6424].

Illustrative attention module. All displayed input maps keep two input locations; no model training or image selection.